

PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

Pentester Name	[Your Name] (Cybersecurity Professional)
Program/Batch	Networkwalks
Date	23 September 2026
Modules Completed	W2-PM1 (Multiple Kali Tools) W2-PM (Nmap Scanning)
Client/Target	1. Networkwalks (secured written permission already) 2. My own local network
Permission Secured from Client?	Yes
Phases Covered	Phase 1: Reconnaissance & Footprinting Phase 2: Scanning & Network Discovery Phase 3-5: In Progress

1. Liability Disclaimer

I have performed these activities only on the systems & devices where I had secured written permission or the devices/systems that I own myself. All these materials are for education and research purpose only. Do not use anything from here to break the law. The instructor, the authors and Networkwalks are not responsible for what you do with this knowledge. Misuse can lead to criminal charges, heavy fines, loss of your job and a permanent record. In most countries unauthorised access is a crime even when nothing is damaged.

2. Introduction

This report covers footprinting the networkwalks.com domain using multiple Kali Linux tools and scanning a host on my own local network using Nmap. Together they show how an attacker moves from gathering public information to confirming a live host on a network. It is part of my ongoing internship program at Networkwalks.

All commands were run in Kali Linux. Every step below includes the exact command used, the result I observed, a screenshot as evidence, and a short note on why the finding matters from an attacker's point of view.

3. Tools Used

The table below lists each tool used in this report and its purpose.

Tool	Purpose
Kali Linux	Operating system used for reconnaissance activities
WHOIS	Find domain registration details (owner, dates, name servers)
whatweb	Fingerprint web technologies (server, CMS, plugins, IP)
nslookup	Resolve the domain name to its IP address using DNS
curl -I	Read the HTTP response headers of the website
wafw00f	Detect whether a Web Application Firewall protects the site
Nmap (CLI) / Zenmap	Scan a host on the local network to confirm it is live

4. Activities Performed

4.1 Footprinting & Reconnaissance

I performed reconnaissance against the networkwalks.com domain using five Kali Linux tools: WHOIS, WhatWeb, Nslookup, Curl and Wafw00f. Each tool was used to collect a different type of information about the target.

First, I used WHOIS to obtain publicly available domain registration information and identify the domain's name servers. The domain is registered with GoDaddy.com, LLC, was created on 06 November 2019 and is set to expire on 06 November 2027. The name servers returned were NS6135.HOSTGATOR.COM and NS6136.HOSTGATOR.COM (with NS29/NS30.DOMAINCONTROL.COM also listed by the registrar record), and the registrant's contact details are privacy-protected via Domains By Proxy, LLC.

I then used WhatWeb to identify technologies used by the website. The results identified WordPress 7.1.2 together with WordPress Download Manager 3.3.58, a Bootstrap 7.1.2 front end, jQuery 3.7.1, the server IP address 192.232.216.135, and a public contact address (info@networkwalks.com).

Using Nslookup, I resolved the domain name to its IP address. The result identified 192.232.216.135, matching the IP exposed by WhatWeb.

I used Curl with the -I option to inspect the HTTP response headers. The site returned HTTP/2 200, exposed the WordPress REST API endpoint (/wp-json/) via a Link header, set a client-side cookie (__wpdm_client), and disclosed permissions-policy directives referencing third-party services (Google, reCAPTCHA, hCaptcha, Cloudflare).

Finally, I used Wafw00f to determine whether a Web Application Firewall was protecting the website. The result identified ModSecurity (SpiderLabs), detected using 2 requests.

4.2 Network Scanning with Nmap

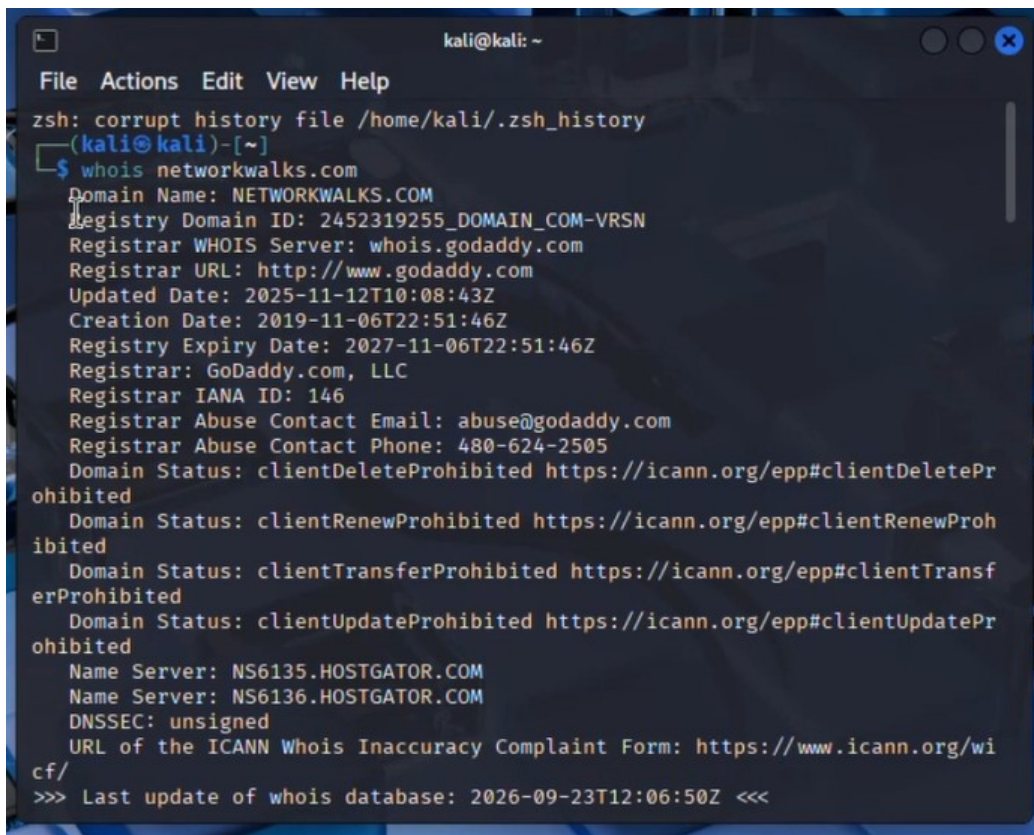
For the second activity, I used Nmap to check whether a host on my local network was live. The scan targeted 192.168.52.1.

Command: `nmap 192.168.52.1`

Nmap 7.991 reported the scan started at 2026-09-23 20:02 (+0500) and confirmed that host 192.168.52.1 is up. The scan completed in 0.83 seconds, covering 1 IP address (1 host up). The accompanying Nmap/Zenmap Host Details view confirms the host state as "up" with 0 open, filtered, closed and scanned ports, meaning this was a host-discovery (ping) check rather than a full port scan — no port, service or MAC address information was returned for this host.

5. Evidences Collected

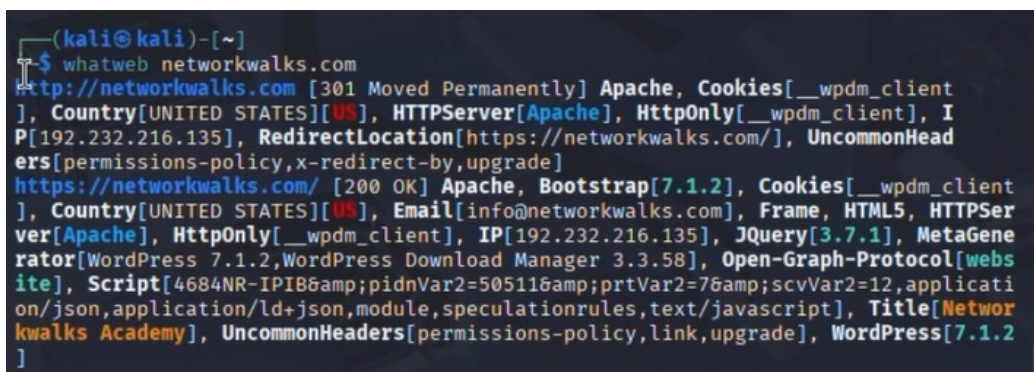
5.1 WHOIS — networkwalks.com

A terminal window titled 'kali@kali: ~' showing a WHOIS lookup for 'networkwalks.com'. The output displays domain registration details including the registrar (GoDaddy.com, LLC), creation and expiry dates, and name servers (NS6135.HOSTGATOR.COM and NS6136.HOSTGATOR.COM). The terminal has a menu bar with 'File', 'Actions', 'Edit', 'View', and 'Help'.

```
kali@kali: ~
File Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history
(kali@kali)-[~]
$ whois networkwalks.com
Domain Name: NETWORKWALKS.COM
Registry Domain ID: 2452319255_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.godaddy.com
Registrar URL: http://www.godaddy.com
Updated Date: 2025-11-12T10:08:43Z
Creation Date: 2019-11-06T22:51:46Z
Registry Expiry Date: 2027-11-06T22:51:46Z
Registrar: GoDaddy.com, LLC
Registrar IANA ID: 146
Registrar Abuse Contact Email: abuse@godaddy.com
Registrar Abuse Contact Phone: 480-624-2505
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeletePr
ohibited
Domain Status: clientRenewProhibited https://icann.org/epp#clientRenewProh
ibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransf
erProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdatePr
ohibited
Name Server: NS6135.HOSTGATOR.COM
Name Server: NS6136.HOSTGATOR.COM
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wi
cf/
>>> Last update of whois database: 2026-09-23T12:06:50Z <<<
```

Figure 1: WHOIS lookup showing registrar, creation/expiry dates and name servers.

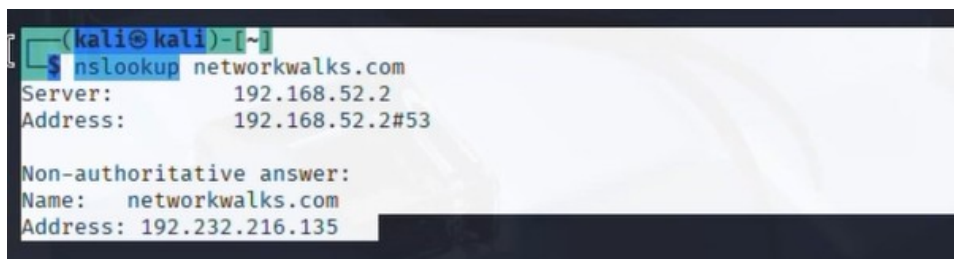
5.2 WhatWeb — networkwalks.com

A terminal window titled '(kali@kali)-[~]' showing a WhatWeb fingerprinting result for 'networkwalks.com'. The output identifies the server as Apache, the CMS as WordPress 7.1.2, and lists various plugins and meta-data. The terminal has a menu bar with 'File', 'Actions', 'Edit', 'View', and 'Help'.

```
(kali@kali)-[~]
$ whatweb networkwalks.com
http://networkwalks.com [301 Moved Permanently] Apache, Cookies[__wpdm_client
], Country[UNITED STATES][US], HTTPServer[Apache], HttpOnly[__wpdm_client], I
P[192.232.216.135], RedirectLocation[https://networkwalks.com/], UncommonHead
ers[permissions-policy,x-redirect-by,upgrade]
https://networkwalks.com/ [200 OK] Apache, Bootstrap[7.1.2], Cookies[__wpdm_client
], Country[UNITED STATES][US], Email[info@networkwalks.com], Frame, HTML5, HTTPSer
ver[Apache], HttpOnly[__wpdm_client], IP[192.232.216.135], JQuery[3.7.1], MetaGene
rator[WordPress 7.1.2,WordPress Download Manager 3.3.58], Open-Graph-Protocol[webs
ite], Script[4684NR-IPIB&pidnVar2=505116&pvtVar2=76&scvVar2=12,applicati
on/json,application/ld+json,module,speculationrules,text/javascript], Title[Networ
kwalks Academy], UncommonHeaders[permissions-policy,link,upgrade], WordPress[7.1.2
]
```

Figure 2: WhatWeb fingerprinting result identifying WordPress, plugins and server IP.

5.3 Nslookup — networkwalks.com

A terminal window titled '(kali@kali)-[~]' showing an Nslookup query for 'networkwalks.com'. The output displays the server IP (192.168.52.2) and the address (192.168.52.2#53). The terminal has a menu bar with 'File', 'Actions', 'Edit', 'View', and 'Help'.

```
(kali@kali)-[~]
$ nslookup networkwalks.com
Server: 192.168.52.2
Address: 192.168.52.2#53

Non-authoritative answer:
Name: networkwalks.com
Address: 192.232.216.135
```

Figure 3: *Nslookup* resolving the domain to its IP address.

5.4 Curl -I — networkwalks.com

```
(kali@kali)-[~]
└─$ curl -I https://networkwalks.com
HTTP/2 200
permissions-policy: private-state-token-redemption=(self "https://www.google.com"
"https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.c
om" "https://hcaptcha.com"), private-state-token-issuance=(self "https://www.googl
e.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloud
flare.com" "https://hcaptcha.com")
link: <https://networkwalks.com/wp-json/>; rel="https://api.w.org/", <https://netw
orkwalks.com/wp-json/wp/v2/pages/53>; rel="alternate"; title="JSON"; type="applica
tion/json", <https://networkwalks.com/>; rel=shortlink
set-cookie: __wpdm_client=9b78681742b03342f7bd00500607b022; path=/; domain=network
walks.com; secure; HttpOnly
content-type: text/html; charset=UTF-8
date: Wed, 23 Sep 2026 12:09:37 GMT
server: Apache
```

Figure 4: HTTP response headers returned by curl -I.

5.5 Wafw00f — networkwalks.com

The screenshot shows a Kali Linux terminal window with the following content:

```
kali@kali: ~  
File Actions Edit View Help  
$ wafwoof networkwalks.com  
wafwoof: command not found  
  
(kali@kali)~  
$ wafw00f networkwalks.com
```

A large ASCII art graphic is displayed, featuring a dog's head profile on the left and various HTTP status codes on the right:

- W00f!
- 404 Hack Not Found
- 405 Not Allowed
- 403 Forbidden
- 502 Bad Gateway
- 500 Internal Error

Below the graphic, the version information is shown:

```
~ WAFW00F : v2.3.1 ~
```

A white box highlights the title of the tool:

The Web Application Firewall Fingerprinting Toolkit

Below the title, the output of the scan is displayed:

```
[*] Checking https://networkwalks.com  
[+] The site https://networkwalks.com is behind ModSecurity (SpiderLabs) WAF.  
[~] Number of requests: 2
```

The terminal prompt at the bottom indicates the user is back in the shell:

```
(kali@kali)~  
$
```

Figure 5: Wafw00f identifying ModSecurity (SpiderLabs) as the WAF in front of the site.

5.6 Nmap / Host Details — 192.168.52.1

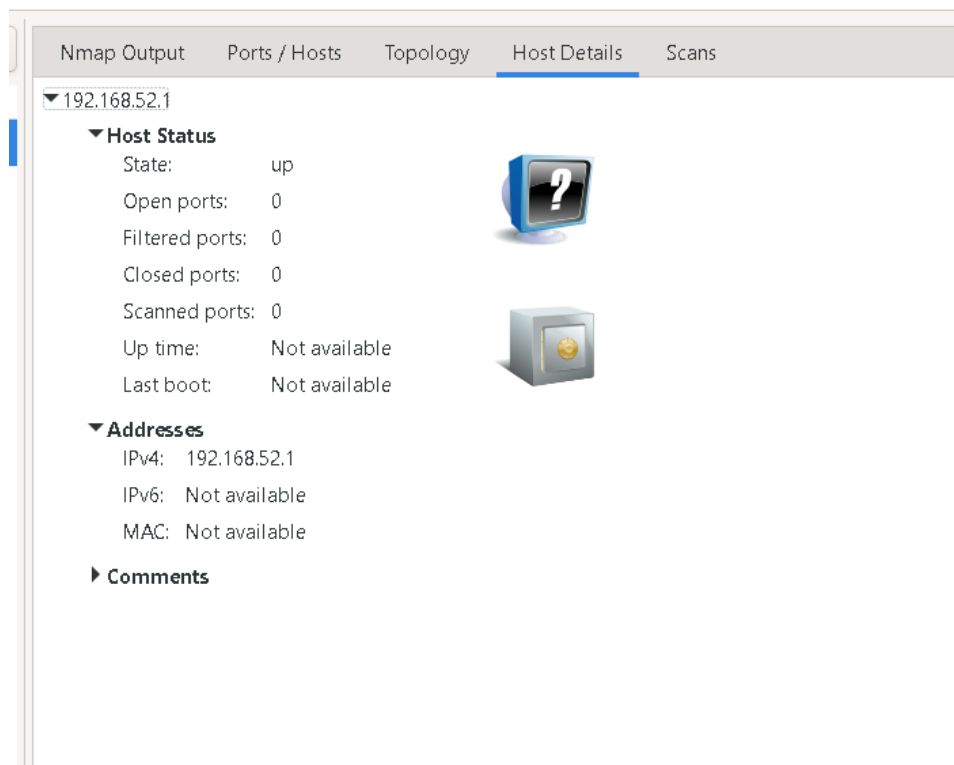


Figure 6: Host Details view confirming 192.168.52.1 is up, with no ports scanned.

6. Risk Analysis / Impact

Based on the information collected during the footprinting and scanning activities, I identified the following potential risks.

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	Web technology information exposed	whatweb identified WordPress 7.1.2 and WP Download Manager 3.3.58	Attackers may use exposed technology/version information to identify software requiring further security review	● Medium
2	Server IP address identifiable	nslookup resolved the domain to 192.232.216.135	Provides information about the network location of the web service	● Low
3	HTTP technical information exposed	curl -I returned HTTP response headers and exposed the /wp-json/ REST API endpoint	May assist technology fingerprinting and further enumeration	● Low
4	WAF technology identifiable	wafw00f identified ModSecurity (SpiderLabs)	Reveals information about the web application's security architecture	● Low
5	Domain registration infrastructure exposed	whois identified registrar (GoDaddy), creation/expiry dates and name servers	Can help build a broader infrastructure and hosting profile of the target	● Low
6	Live host confirmed on local network	Nmap confirmed host 192.168.52.1 is up (no ports scanned)	Confirms a device is active and reachable; further port/service scanning would reveal more	● Low

Risk level key: ● Critical ● Medium ● Low

The risks above are observations from the footprinting and scanning exercises, not confirmed vulnerabilities. The practical exercises primarily involved information gathering and host discovery — no exploitation or vulnerability validation was performed. Therefore, the presence of information such as a software version, IP address or DNS record does not by itself mean that the system is vulnerable. Further authorized security testing would be required to confirm any actual vulnerability.

7. Recommendations

- Review publicly exposed technology information — organizations should regularly review what information about their web technologies, CMS and plugins is publicly visible.
- Keep software updated — CMS platforms, plugins and other web technologies should be regularly updated and reviewed against current security advisories.
- Review HTTP headers — HTTP response headers should be reviewed to determine whether unnecessary technical information is being exposed.
- Review WHOIS/domain exposure — confirm registrar-level privacy protection stays enabled and name-server configuration is reviewed periodically.
- Properly configure and monitor the WAF — keep the WAF (ModSecurity) enabled and tuned, since it already blocks naive attacks.
- Perform regular internal network discovery — organizations should periodically scan their own networks to identify active devices.
- Follow up with a full port/service scan — since this scan only confirmed the host was up, a full port and service scan should be performed (with authorization) to identify running services.
- Perform security testing with authorization — reconnaissance and scanning should only be performed against systems and networks where appropriate authorization has been provided.

8. Conclusion

During this phase of my Cybersecurity & Ethical Hacking internship, I completed practical activities covering footprinting, reconnaissance and network scanning.

In the footprinting activity, I used five Kali Linux tools to collect information about the target domain. I learned how WHOIS can provide domain information, WhatWeb can identify web technologies, Nslookup can resolve domain names, Curl can inspect HTTP headers, and Wafw00f can identify a WAF.

In the network scanning activity, I used Nmap to confirm that a host on my local network was live and reachable.

The exercises showed me that information gathering is an important part of cybersecurity. Even before attempting to exploit a system, a security professional can learn a significant amount about an environment by carefully analyzing publicly available information and network responses.

Finally, I learned that reconnaissance and scanning must always be performed within an authorized scope. These activities were completed as part of the assigned educational cybersecurity lab.